



YOUCHENG LAN

Prospective RA Applicant
Multi-Agent Reinforcement Learning, AI for
Economics, and Financial Agents
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EDUCATION

Shandong University

Second-year Undergraduate Student, Dual-Degree Track

Current

Jinan, China

Academic preparation: multi-agent reinforcement learning, LLM-based agents, AI for economics, financial-agent benchmarks, market mechanism simulation, and empirical evaluation.

RESEARCH EXPERIENCE

Center for Artificial Intelligence and Digital Finance, School of Management, Shandong University

Current

Undergraduate Research Assistant

Jinan, China

- Conduct hands-on research in MARL, LLM-based agents, multi-agent market simulation, agent evaluation, trading benchmarks, and financial AI.
- Work on literature review, code reproduction, data construction, experiment implementation, model evaluation, result validation, and manuscript writing.

RESEARCH FIT

My current research uses financial and on-chain markets as executable multi-agent environments to study agent coordination, role specialization, mechanism simulation, and reliable evaluation. This background connects naturally with multi-agent reinforcement learning and AI for economics.

SELECTED WORKING PAPERS AND PROJECTS

Language-Guided Multi-Agent Reinforcement Learning for Robust Arbitrage-Aware Market Making in Decentralized Exchanges

AAAI 2027 targeted submission; first-author working paper (completed draft)

Studies how language-derived market beliefs can guide multi-agent market-making policies under arbitrage opportunities, liquidity shocks, gas changes, and regime shifts.

My work: Built the calibrated DEX simulator interface, rule baselines, market-summary and belief-label schemas, student-belief distillation workflow, MAPPO-style smoke tests, robustness tests, and market-quality/risk metrics.

DEXTwin: A Data-Calibrated LLM Agent Simulator for Mechanism Design in Decentralized Exchanges

AAAI 2027 targeted submission; first-author working paper (completed draft)

Builds a data-calibrated multi-agent economic simulator with heterogeneous traders, arbitrageurs, and liquidity providers for mechanism and market-behavior studies.

My work: Integrated CEX prices, DEX pool/reserve data, gas data, and news context; implemented the Uniswap V2-style simulator, agent action schema, validation metrics, and mechanism counterfactual setup.

RiskArena: A Regime-Stratified Benchmark for Position-Aware Multi-Agent Trading

AAAI 2027 targeted submission; first-author working paper (completed draft)

Frames financial-agent evaluation as an executable multi-agent portfolio task with regime-conditioned scorecards, transaction frictions, and reproducible protocols.

My work: Constructed the S&P 100 public-data benchmark stack, regime labels, transaction-cost/slippage/liquidity execution layer, LLM-agent adapters, deterministic baselines, score diagnostics, cost ablations, and reproducibility materials.

TracePnL: Counterfactual Provenance Attribution for Portfolio Decisions

AAAI 2027 targeted submission; first-author working paper (completed draft)

Develops decision-trace attribution tools that connect realized trading outcomes to evidence, memory, debate, and agent-role traces.

My work: Implemented the provenance graph logger, counterfactual replay engine, Trading Attribution Score, Causal Rationale Gap labels, surrogate replay model, role/news ablations, stability analysis, and expert case-study pack.

TECHNICAL BACKGROUND

- Python-based experiment pipelines, data processing, financial backtesting, and reproducible empirical workflows.
- Reinforcement learning and multi-agent experiment frameworks, benchmark construction, model evaluation, and robustness testing.
- LLM-based agents, decision traces, market simulation, information-time controls, and manuscript-ready result organization.

TESTS, QUALIFICATIONS, AND BACKGROUND

IELTS: 6.5

Former Modern Pentathlon Athlete, Shenzhen National First-Class Swimmer, China ABRSM Piano Grade 8